

MATH 1342 – Elementary Statistics – Week 13: Inference for Two Means and Paired Data

Nov 16 – 18 | 6.4 Inference for a Difference in Means (Distribution, CI, Hypothesis Test) • 6.5 Paired Difference in Means | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 7. Apply ideas of appropriate sampling techniques and experimental design to data production. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Compute the standard error and conservative degrees of freedom for a difference in two independent means, and check the conditions. (CO4, CO8) 2. Construct and interpret a confidence interval for $\mu_1 - \mu_2$, explaining how the order of the groups affects the sign. (CO9) 3. Carry out a two-sample t test, choosing the tail from the research question. (CO10) 4. Classify a study design as paired or independent samples by checking for a link between observations. (CO7, CO10) 5. Define the difference variable d, carry out a paired t test and paired t confidence interval, and explain why pairing reduces variability. (CO9, CO10)	e-Text: 6.4-CI Confidence Interval for a Difference in Means	MO1, MO2, MO3	Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test	MO1, MO2, MO3	Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM)	MO1, MO2, MO3
		e-Text: 6.4-D Distribution of a Difference in Means	MO1, MO2, MO3	Participation: Paired or Independent? – class discussion of eight study scenarios (a)–(h); define d and the order of subtraction	MO4, MO5	Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM)	MO4, MO5
		e-Text: 6.4-HT Hypothesis Test for a Difference in Means	MO1, MO2, MO3				
		Interactive Lecture: 6.4 Inference for a Difference in Means	MO1, MO2, MO3			Test 3 – Fri Nov 20 – Sun Nov 22 (Blackboard, proctored, 2 hours). Covers Weeks 10–13.	All
		Slides: 6.4 Inference for a Difference in Means	MO1, MO2, MO3				
		e-Text: 6.5 Paired Difference in Means	MO4, MO5				
		Interactive Lecture: 6.5 Paired Difference in Means	MO4, MO5				
		Slides: 6.5 Paired Difference in Means	MO4, MO5				
						Take-Home Quiz 8 (due Mon Nov 23, 11:59 PM)	All